

OCEANA NAS, VA, USA

WMO: 723075

Lat: 36.817N

Lon: 76.033W

Elev: 7

StdP: 101.24

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13769

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-3.8	-15.9	0.9	-3.2	-13.2	1.2	-1.5	13.0	5.4	11.7	5.0	4.5	340	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	33.9	25.1	32.6	24.6	31.3	24.1	26.3	31.9	25.6	30.7	25.1	29.7	4.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	19.8	28.9	24.1	19.0	28.1	23.7	18.6	27.7	81.6	31.9	78.6	30.7	76.2	29.7	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.5	8.5	-9.4	36.4	2.8	1.3	-11.4	37.3	-13.0	38.1	-14.5	38.9	-16.5	39.8		
(5)			-10.5	27.7	2.6	0.8	-12.3	28.3	-13.8	28.8	-15.3	29.3	-17.1	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	5.2	6.4	9.8	15.0	19.5	24.1	26.3	25.5	22.6	17.1	11.2	7.5
	DBStd	8.62	5.82	5.28	5.29	4.85	4.16	3.34	2.57	2.49	2.89	4.28	4.78	5.31
	HDD10.0	532	171	126	71	9	0	0	0	0	0	2	41	113
	HDD18.3	1811	407	334	269	124	39	3	0	0	3	76	219	336
	CDD10.0	2688	23	26	65	159	293	422	506	482	377	223	76	37
	CDD18.3	925	1	1	5	24	74	175	247	223	130	38	4	2
	CDH23.3	7761	1	6	44	201	550	1522	2427	1982	800	202	17	8
	CDH26.7	2605	0	0	6	44	159	543	946	683	187	34	1	1
Wind	WSAvg	4.1	4.5	4.5	4.7	4.8	4.2	3.8	3.5	3.3	3.8	3.9	4.0	4.1
Precipitation	PrecAvg	1123	96	87	95	70	89	90	128	128	98	86	65	77
	PrecMax	1479	289	228	207	167	168	190	315	346	388	254	144	148
	PrecMin	713	31	17	23	25	11	21	24	36	5	5	2	15
	PrecStd	173	52	41	42	35	40	42	64	75	80	62	38	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	24.1	27.4	30.1	32.4	34.9	36.3	35.4	32.4	29.4	25.3	24.4
		MCWB	16.9	16.9	17.7	19.6	22.5	24.6	26.4	25.9	24.1	21.9	19.1	19.1
	2%	DB	19.7	20.9	24.1	27.9	30.3	32.9	34.1	33.2	30.5	27.5	22.8	21.2
		MCWB	15.8	14.9	16.5	19.1	22.0	24.0	25.4	25.2	23.5	21.4	18.0	17.3
	5%	DB	17.3	18.2	21.5	25.9	28.6	31.4	32.8	31.9	28.8	25.6	21.0	18.8
		MCWB	14.0	14.0	15.0	18.0	21.4	23.6	25.2	24.7	22.9	20.7	17.2	15.9
	10%	DB	14.9	15.6	18.8	23.2	26.7	29.8	31.4	30.2	27.3	23.8	18.9	16.4
		MCWB	11.9	11.7	14.5	17.2	20.3	23.1	24.6	24.0	22.3	19.8	15.6	13.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	18.4	19.4	21.9	24.1	25.9	27.5	27.2	25.5	24.1	21.2	20.2
		MCDB	21.1	22.5	23.8	26.7	29.6	31.8	33.7	32.9	29.0	26.9	23.5	22.8
	2%	WB	16.5	16.8	18.0	20.6	23.1	25.2	26.6	26.2	24.8	22.9	19.4	18.1
		MCDB	19.1	19.8	22.0	25.6	28.1	30.6	32.5	31.4	28.1	25.7	21.7	20.5
	5%	WB	14.4	14.6	16.7	19.4	22.2	24.6	25.8	25.5	24.2	21.8	17.8	16.3
		MCDB	17.0	17.9	20.7	24.1	26.7	29.6	31.2	30.1	27.2	24.6	20.1	18.4
	10%	WB	12.1	11.8	14.7	18.0	21.3	24.0	25.2	24.9	23.6	20.5	16.3	14.1
		MCDB	14.5	15.0	18.4	22.2	25.5	28.5	30.2	29.0	26.3	23.3	18.6	16.4
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.7	9.8	10.3	9.3	8.5	8.3	7.9	7.8	9.2	9.7	9.3
		MCDBR	12.3	13.7	13.8	13.1	11.6	10.4	9.9	9.6	9.5	10.3	11.3	11.3
		MCWBR	9.5	9.7	8.2	6.5	5.5	4.2	4.0	3.9	4.4	5.5	7.7	8.8
	5% WB	MCDBR	11.4	12.7	12.1	11.5	10.1	9.1	9.0	8.4	7.4	8.2	9.7	10.7
		MCWBR	9.7	10.4	9.0	7.0	5.4	4.2	4.2	3.9	3.8	4.9	7.9	9.4
Clear-Sky Solar Irradiance	taub	0.312	0.323	0.349	0.392	0.437	0.483	0.511	0.504	0.409	0.358	0.327	0.316	
	taud	2.542	2.505	2.463	2.353	2.261	2.181	2.118	2.125	2.384	2.517	2.551	2.559	
	Ebn at Noon	885	916	918	890	851	808	783	779	844	864	860	856	
	Edh at Noon	78	91	104	122	136	147	156	152	111	89	77	72	
All-Sky Solar Radiation	RadAvg	2.32	3.10	4.16	5.31	5.99	6.41	6.11	5.51	4.51	3.56	2.63	2.08	
	RadStd	0.19	0.32	0.32	0.38	0.50	0.41	0.29	0.33	0.35	0.38	0.21	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.43	N/A	-1.20	N/A	N/A	N/A	N/A	N/A	+119	+91
(47) Regional (52 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	+105

Nomenclature: See separate page